

AMIT ROY

Cybersecurity Engineer · Ethical Hacker · Penetration Tester · IoT & AI Security Specialist

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PROFESSIONAL SUMMARY

CEH-certified Ethical Hacker and Cybersecurity Engineer with 1+ years of hands-on experience across Web Application & Network Penetration Testing, reverse engineering, malware analysis, AI-driven security tooling, and IoT/embedded systems security. Self-taught polyglot programmer in 28+ languages with a strong mathematical foundation — excelling in algorithm design, cryptography, and AI/ML modelling. Builder of open-source security tools (SIEM, SOC tooling, phishing detection, malware AI). First-Prize CTF winner. Passionate about offensive security, IoT/electronics, and AI-integrated threat defence — backed by a structured 4-year Honours syllabus spanning networking, forensics, cloud, and AI in cybersecurity.

TECHNICAL SKILLS

Security Domains

Web App Pentesting (OWASP Top 10) · Network Pentesting · Reverse Engineering · Malware Analysis · Digital Forensics · SOC Operations · Cryptography · Vulnerability Analysis · Social Engineering & OSINT · Steganography · Blockchain Security · AI Security · Cyber Laws

Cybersecurity & Offensive Tools

Metasploit · Burp Suite · Nmap · Wireshark · OWASP ZAP · Aircrack-ng · Nikto · SQLMap · Hydra · John the Ripper · Ghidra · IDA Pro · Volatility · Snort · Splunk · YARA Rules · Kali Linux · ParrotOS

Networking & Infrastructure

TCP/IP · Switching & Routing (CCNA) · Linux Administration · Windows Server 2019/2022 · Firewall & IDS/IPS · VPN · DNS/DHCP · Linux Network Services

Cloud & Virtualisation

Cloud Security · Virtualisation Technology · Windows Server Management · Security Audit & Compliance · Digital Watermarking

AI/ML & Security

Machine Learning · Neural Networks · NLP for Threat Detection · Anomaly Detection · AI-powered Malware Classification · TensorFlow · scikit-learn · Adversarial Machine Learning · Prompt Injection Defense · Model Poisoning Detection · LLM Security · AI Red Teaming · Model Evasion Techniques

Primary Programming Languages

Python · Java · Go · C · C++ · JavaScript · TypeScript · Assembly · Kotlin · Rust

Scripting Languages

Bash · PowerShell · Batch Script · VBScript

Additional Languages

PHP · Ruby · Perl · Lua · Lisp · C# (.NET) · R · MATLAB · Zig · Swift · Nim · Scala · Objective-C · HTML · CSS

Frameworks & Dev Tools

React · Django · Flask · Bootstrap · Tailwind CSS · Node.js · Next.js · Vue.js · Vite · jQuery · Git & GitHub · VS Code

Architecture & Design

Cybersecurity System Architecture · System Design · Network Architecture · Zero-Trust Design · OS Internals (Linux & Windows) · Secure SDLC · DSA & Algorithm Design · Mathematics (Strong Foundation — Cryptography, AI/ML, DSA)

DevSecOps / DevOps

CI/CD Security (GitHub Actions, Jenkins) · SAST / DAST Integration · Container Security (Docker) · Infrastructure as Code Security · Shift-Left Security · Vulnerability Management Pipelines · Secure Software Development Lifecycle (SSDLC) · YAML/YML Configuration · Ansible · Kubernetes · XML · JSON

IoT , Electronics & Microcontrollers

Arduino · ESP32 · MSP430 · 8051 · RP2040 · MSP430 · PIC · nRF52 · LPC / i.MX RT · ATmega328P · RP2040 · STM32 · Raspberry Pi · RFID/NFC Systems · IoT Firmware Analysis & Exploitation · IoT Security · Sensor Networks · Fundamentals of Electronics · Embedded C & Assembly · Circuit Design · VLSI Chip Designing · FPGA

ACADEMIC SKILL COVERAGE — BSC ANCS 4-YEAR SYLLABUS

Key domains across all 8 semesters — mapped to industry-relevant skills:

Sem 1 • Foundations

Computer Networks · PC Maintenance · Fundamentals of Electronics · Computer Peripherals & OS

Sem 2 • Linux & Security Basics

Linux Administration · Cyber Security Lab · Principles of Programming · Security Overview

Sem 3 • Network Security & Pentesting

Switching & Routing · Vulnerability Analysis & Penetration Testing · Information Security Management

Sem 4 • Web Security & Databases

Windows Server Administration · Web Application Penetration Testing · Database Management System

Sem 5 • Advanced Networking

Network Security Essentials · Linux Network Service Management · Blockchain & Cryptocurrency · Digital Watermarking & Steganography

Sem 6 • AI/ML & SOC

Windows Server Management · Security Operations Center Management · AI & Machine Learning · Cryptography & Network Security

Sem 7 • Cloud, Forensics & IoT (Hons)

Security Audit & Compliances · Cloud Security · Digital Forensics · IoT Security

Sem 8 • AI Security & Cyber Law (Hons)

Data Security & Analytics · AI in Cyber Security · Virtualisation Technology · Cyber Security Law & IPR

FEATURED PROJECTS — GITHUB.COM/DEDSECHACK-1337

soc-copilot · Python · LangChain · Ollama · RAG

- ▶ Local-first AI Threat Hunting Assistant that translates analyst natural-language queries into Sigma and YARA detection rules, auto-validated via pySigma and yara-python before returning results.
- ▶ RAG-powered MITRE ATT&CK mapping (697 techniques via ChromaDB + Ollama embeddings) driven by a LangChain/LangGraph tool-calling agent with a Streamlit chat interface.

PhoenixSIEM · TypeScript · Self-Learning AI · SOC

- ▶ Self-learning SIEM platform that ingests multi-source logs, detects anomalies, and surfaces prioritised alerts — cutting SOC alert fatigue by auto-filtering false positives.
- ▶ Bridges SOC Management, AI in Cyber Security, and Virtualisation Technology into a full-stack production security product.

AI-NIDPS (Sentry) · Python · Scapy · ML/DL · Network Security

- ▶ AI-powered Network Intrusion Detection System with live packet capture — classifies DoS, DDoS, Port Scan, Brute Force & Botnet traffic via RandomForest, XGBoost, LightGBM, and a PyTorch deep learning model.
- ▶ Flask-based SOC-style dashboard with live flow logs, attack breakdown charts, and PDF report export — staged to upgrade from synthetic data to real CICIDS2017 / UNSW-NB15 datasets without a rewrite.

malware-guardAI · Python · AI/ML · Malware Analysis

- ▶ AI-powered engine that classifies and dissects malicious binaries using ML, delivering automated threat reports — reduces manual analysis effort by automating static and dynamic pipelines.
- ▶ Directly applies AI in Cyber Security and Data Security & Analytics coursework to a real-world production security tool.

PhishguardAI · Python · NLP · Threat Intelligence

- ▶ Real-time phishing URL and email detector using NLP and heuristics to catch zero-day campaigns before credential exposure occurs.

- Applies AI in Cyber Security and Cryptography & Network Security — demonstrates end-to-end AI-driven threat intelligence pipeline.

network_scanner · Python · Network Reconnaissance

- Full-featured subnet/host scanner — fingerprints services, detects open ports and OS signatures, outputs structured JSON for SIEM ingestion.
- Applies Data Security & Analytics and Network Security Essentials to automate real-world network reconnaissance workflows.

RFID Cloner · Arduino · C · Hardware / IoT Security · Final Year College Project

- Custom-built RFID cloning device (125kHz EM4100 & MIFARE Classic) demonstrating physical access control vulnerabilities — used in live security awareness demos.
- Directly reflects IoT Security and Cyber Security Law & IPR — built to demonstrate real-world hardware attack surfaces in enterprise environments.

WORK EXPERIENCE

Cybersecurity Intern — Ethical Hacking Division

2019 – 2020

Indian Cyber Security Solutions | ISO 27001 & 9001:2015

- Performed web application pentests (OWASP Top 10) and network pentests using Metasploit & Nmap — produced risk-rated vulnerability reports with remediation roadmaps.
- Won 1st Prize in the company CTF Competition (Aug 2022); earned CEHP and CCNA through their ISO-accredited programme.

IT Cell Intern — Cybersecurity

2024 (6 Months)

Brainware University, Kolkata

- Conducted internal vulnerability assessments; deployed Wireshark & Snort for real-time anomaly detection.
- Delivered security awareness workshops to 100+ staff on phishing, social engineering, and incident response.

Earn While You Learn — Technical Staff

Apr – Aug 2024

Brainware University | Ref: BWU/REG/ELS/STD/01042024/08

- Competitively selected from a limited cohort for paid technical support role within the university IT department — performance evaluated monthly by department supervisors.

EDUCATION

BSc (H) — Advanced Networking & Cyber Security (ANCS)

2023 – Present

Brainware University, Barasat, Kolkata | Roll: 23010348145 | Reg: 23013003403

6th Sem SGPA: **9.50 (90.50%)** · Overall till 6th Sem: **82.38%** · Specialisations: AI/ML · Cryptography · SOC Management · Windows Server · Cloud Security · Digital Forensics · IoT Security

Higher Secondary — Engineering & Technology (Vocational)

2022 – 2023

Ashoknagar Boys Secondary School | WB State Council of Technical & Vocational Education | Roll: 221079114013

79.60% · Grade A · Discipline: Engineering & Technology (ETCM)

Secondary Examination (Madhyamik Pariksha)

2014 – 2015

Ashoknagar Boys Secondary School | WBBSE | Roll: 204301B

457 / 700 · Overall Grade: A

CERTIFICATIONS & LICENCES

CEHP

Certified Ethical Hacking Professional
No. ICSS369034002019
Nov 2019 · Indian Cyber Security Solutions

CCNA (Grade A)

Cisco Certified Network Associate
No. ICSS36903292019
Feb 2019 · Indian Cyber Security Solutions

Practical Ethical Hacking

Complete Course — TCM Security
11 Nov 2020

Ethical Hacking — O+

Industrial Training · ISOEH (EC-Council)
Reg: ISOEH/EH/24/1670
Jul–Aug 2024

Network Pen Testing — O+

Level-1 · ISOEH (EC-Council Accredited)
Reg: ISOEH/NP/25/1843
Jan 2025

IT-SMART — Distinction

APLL Learning Solutions
Cert No. 1790437
Oct 2017

MeitY — ISEA

Cyber Hygiene Practices · Govt of India
No. MeitY/ISEA/WCHP/023098
20 Oct 2022

TryHackMe — AoC 2025

Advent of Cyber · 24 Challenges Completed
THM-AQA6PIKKLL
7 Jan 2026

Ethical Hacking — Megamind

Certificate of Participation · Nikistian Media
8 Jan 2023

Getting Started with AI

IBM SkillsBuild · Credly Badge
PLAN-E624C2604060
10 Aug 2025

Journey to Cloud

Envisioning Your Solution · IBM SkillsBuild
Badge: 7a4e22af-2136-4b8f-9e44-e30b065f34a
16 Aug 2025

AI Doc Retrieval Lab

IBM Granite & Docling · Adobe Learning Manager
ALM-COURSE_3946473
16 Aug 2025 · 30 mins

Web Hacking / Pen Testing

Udemy · Zaid Sabih, zSecurity
UC-694a5e76-b171-4470-8f46-6bed8e8596d5
7 Aug 2022 · 10.5 hrs

COMPETITIONS & ACHIEVEMENTS

- ★ **1st Prize — CTF Competition** · Indian Cyber Security Solutions · 14 Aug 2022
 - Won first place in a competitive CTF among professional cybersecurity practitioners — demonstrated elite real-world offensive security skills.
- ◆ **Kolkata Police & IEM Labs Hackathon 2022** · Netaji Indoor Stadium, Kolkata · 29 Jul 2022
 - Competed in city-level hackathon co-organised by Kolkata Police and IEM Labs — cybersecurity and software development challenges.
- ◆ **BrainCon 2024 — TechFusion** · Brainware University (IIC, MoE) · 18 May 2024
 - Selected and recognised for commendable performance at Brainware University Ministry of Education Innovation Council event.
- ◆ **Brainware-CTF (CyberYami / BCC)** · Brainware Cyber Club · 30 Nov 2023
 - Earned Certificate of Appreciation from CyberYami & Brainware Cyber Club for BRAINWARE-CTF event.
- ▶ TryHackMe Advent of Cyber 2025 — completed 24 challenges across AI, web, forensics, network analysis (Cert: THM-AQA6PIKKLL).
- ▶ 33 public repositories on GitHub; 5+ actively maintained open-source cybersecurity tools with community recognition.
- ▶ Self-taught in 28+ programming languages — avg. 21–30 days per language; spans systems, data science, and mobile domains.
- ▶ Selected for Brainware University Earn While You Learn Scheme (2024) — limited merit cohort for academic & technical excellence.

COLLEGE ACTIVITIES & PARTICIPATION — BRAINWARE UNIVERSITY

SecCom 2025

Expert Talk · Participation Certificate
31 Oct – 1 Nov 2025

Freshers 2025

Volunteer Certificate · Celebration
13 Oct – 29 Nov 2025

HR Spark 2025

Seminar · Participation Certificate
8 Feb 2025

SecCom 2024 Celebration · Participation Certificate 28–29 Nov 2024	Freshers Welcome 2024 Celebration · Participation Certificate 1–7 Oct 2024	TechFusion 2024 (BrainCon) Intercollege Tech Challenge · Participation 18 May 2024
Annual Sports 2024 Cricket Tournament (Boys) · Participation 27 Feb – 20 Apr 2024	National Science Day Workshop — Penetration Testing & Bug Bounty 29 Feb 2024	SecCom 2023 — Seminar Computer Security Day · Celebration-Seminar 30 Nov – 1 Dec 2023
SecCom 2023 — Competition Computer Security Day · Competition 30 Nov – 1 Dec 2023	Linux Administration Trends Seminar · Modern Trends in Linux Administration 19 Aug 2023	

SOFT SKILLS

Analytical Thinking	Rapid Self-Learning	Problem Solving	Communication	Adaptability
Decision Making	Multi-tasking	Attention to Detail	Team Collaboration	Creative Thinking

INTERESTS & HOBBIES

 Guitar & Music Self-taught guitarist sharing performances on YouTube (@arproduction8597).	 EDM Production & DJ Composes original EDM tracks and mixes using DAWs; passionate about music synthesis and live mixing.	 Electronics & IoT Tinkering Builds custom hardware with Arduino,STM32,nRF52,ESP32 & Raspberry Pi — RFID tools, sensor networks, and embedded systems.
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LANGUAGES

Bengali (Native) · English (Professional) · Hindi (Conversational)